Solution 1

Problem 1.1 — Let $X = (X_1, \ldots, X_n)$ be a random variable, with all components in $L^2(P)$. The covariance matrix $\operatorname{var}(X) = (c_{i,j} : 1 \le i, j \le n)$ of X is defined by $c_{i,j} = \operatorname{cov}(X_i, X_j)$. Show that $\operatorname{var}(X)$ is a non-negative definite matrix.

Well, for any particular vector $v = (v_1, \ldots, v_n)$, we have

$$\begin{aligned}
v^T \operatorname{var}(X) v &= \sum_{i,j} v_i v_j c_{i,j} \\
&= \sum_{i,j} v_i v_j \operatorname{cov}(X_i, X_j) \\
&= \operatorname{cov}\left(\sum_i v_i X_i, \sum_j v_j X_j \right) \\
&= \operatorname{var}\left(\sum_i v_i X_i \right) \\
&\ge 0
\end{aligned}$$

The third line is true because covariance is linear in each random variable; this is clear by starting at the definition. The fourth line is true by definition of variance. The fifth line is true because squares of real numbers are nonnegative, and the integral of a non-negative function is non-negative.

Solution 2

Problem 2.1 — For a finite Borel measure μ on $\mathbb{R}$ show that, if $\int_{\mathbb{R}} |x|^k \, d\mu(x) < \infty$ for $k \in \mathbb{N}$, then the Fourier transform $\hat{\mu}$ of μ has a kth continuous derivative, which at 0 is given by

$$\hat{\mu}^{(k)}(0) = i^k \int_{\mathbb{R}} x^k d\mu(x)$$

First off, note that on finite measure spaces, $L^p \subset L^q$ whenever $p \leq q$ by Holder's inequality;

$$\begin{aligned}
\int_{\mathbb{R}} |x|^p \, d\mu(x) &\leq \left(\int_{\mathbb{R}} |x|^q \, d\mu(x) \right)^{p/q} \left(\int_{\mathbb{R}} 1 d\mu(x) \right)^{1-p/q} \\
&= \left(\int_{\mathbb{R}} |x|^q \, d\mu(x) \right)^{p/q} \mu(\mathbb{R})^{1-p/q} \\
&< \infty
\end{aligned}$$

So we also know that $\int_{\mathbb{R}} |x|^m \, d\mu(x) < \infty$ for all $m \leq k$.

Now, let's use induction to show that

$$\hat{\mu}^{(m)}(t) = i^m \int_E x^m e^{itx} d\mu(x) \tag{$\star$}$$

for all $m \leq k$; this clearly suffices (we know $\hat{\mu}^{(m)}$ is continuous because Fourier transforms are continuous, and we can plug in 0 for the answer). The base case is

$$\hat{\mu}(t) = \int_{\mathbb{R}} e^{itx} d\mu(x)$$

which is true by definition.

Let $f_m(t, x) = i^m x^m e^{itx}$ be the desired integrand. Let $E = \mathbb{R}$ for clarity. Observe

$$\frac{\partial}{\partial t} f_m(t, x) = f_{m+1}(t, x).$$

For all $m \geq 0$, we satisfy:

- For all t, we can integrate $f_m(t, \bullet)$ over E. Indeed,
$$\int_E |f_m(t, x)| \, d\mu(x) \leq \int_E |x|^k \, d\mu(x) < \infty.$$
- For all x, we can differentiate $f_m(\bullet, x)$ with respect to t by construction.
- For all $x \in E$ and all $t \in U$,
$$\left| \frac{\partial}{\partial t} f_m(t, x) \right| \leq |f_{k+1}(t, x)|$$
(in fact they are equal) and the RHS is integrable over E.

Thus, theorem 3.5.1 applies, and we can swap the derivative with the integral; for all m,

$$\begin{aligned}
\frac{\partial}{\partial t} \hat{\mu}^{(m)}(t) &= \frac{\partial}{\partial t} \int_E f_m(t, x) d\mu(x) \\
&= \int_E \frac{\partial}{\partial t} f_m(t, x) d\mu(x) \\
&= \int_E i^{m+1} f_{m+1}(t, x) d\mu(x)
\end{aligned}$$

as desired.

Solution 3

Problem 3.1 — Show that for any real numbers a, b one has $\int_a^b e^{itx}dx \to 0$ as $|t| \to \infty$.

We can just compute:

$$\begin{aligned}\left|\int_a^b e^{itx}dx\right| &= \left|\frac{e^{itx}}{it}\bigg|_a^b\right| \\ &= \left|\frac{e^{itb} - e^{ita}}{it}\right| \\ &= \left|\frac{e^{itb} - e^{ita}}{t}\right| \\ &\leq \frac{2}{|t|} \to 0\end{aligned}$$

where the 2 is by triangle inequality or whatever.

Problem 3.2 — Show that, for any $f \in L^1(\mathbb{R})$, the Fourier transform $\hat{f}(t) = \int_{-\infty}^{\infty} e^{itx}f(x)dx$ tends to 0 as $|t| \to \infty$.

By splitting f into its positive and negative components, WLOG f is nonnegative and integral $\int_{\mathbb{R}} f = M < \infty$. By a well-known lemma, there exists an increasing sequence of nonnegative, simple functions f_n such that $f_n \to f$ pointwise; fix this sequence; by monotone convergence theorem or whatever you want,

$$\int_{\mathbb{R}} f(x) - f_n(x)dx \to 0$$

In particular, for any ε, I can find an N such that $\int_{\mathbb{R}} f(x) - f_N(x) < 0.0001\varepsilon$.

Then, for *any* $t \in \mathbb{R}$,

$$\left|\int_{\mathbb{R}} e^{itx}f(x)dx - \int_{\mathbb{R}} e^{itx}f_N(x)dx\right| = \left|\int_{\mathbb{R}} e^{itx}(f(x) - f_N(x))dx\right| \leq \int_{\mathbb{R}} f(x) - f_N(x)dx \leq 0.0001\varepsilon$$

Now, look carefully at

$$f_N = \sum_{k=1}^{K} c_k \mathbb{1}(E_k), \qquad \int_{\mathbb{R}} f_N \leq \int_{\mathbb{R}} f \implies \sum_k c_k \mu(E_k) \leq M$$

In particular, each E_k has finite, positive measure. By Littlewood's first principle, for each E_k, there is a finite collection of disjoint open intervals $U_k = \bigcup_{n=1}^{N_k}(a_{k,n}, b_{k,n})_{n=1}^{N_k}$ such that

$$\mu(E_k \Delta U_k) < 0.0001\varepsilon/(K \cdot c_k)$$

I.e. we can approximate f_N with

$$g_N = \sum_{k=1}^{K} c_k \sum_{n=1}^{N_k} \mathbb{1}(a_{k,n}, b_{k,n})$$

Great, this means we get the following ridiculous bound:

$$\begin{aligned}
\left|\int_{\mathbb{R}} e^{itx} f_N - \int_{\mathbb{R}} e^{itx} g_N\right| &\leq \left|\sum_{k=1}^{K} \int_{\mathbb{R}} e^{itx} c_k \mathbb{1}(E_k) - \sum_{k=1}^{K} \int_{\mathbb{R}} e^{itx} c_k \mathbb{1}(U_k)\right| \\
&\leq \sum_{k=1}^{K} \left|\int_{E_k \setminus U_k} e^{itx} c_k - \int_{U_k \setminus E_k} e^{itx} c_k\right| \\
&\leq \sum_{k=1}^{K} \left|\int_{E_k \setminus U_k} e^{itx} c_k\right| + \left|\int_{U_k \setminus E_k} e^{itx} c_k\right| \\
&\leq \sum_{k=1}^{K} \left|\int_{E_k \Delta U_k} e^{itx} c_k\right| \\
&\leq \sum_{k=1}^{K} |c_k| \left(\frac{0.0001\varepsilon}{K \cdot c_k}\right) \\
&= 0.0001\varepsilon
\end{aligned}$$

Finally, note that

$$\int_{\mathbb{R}} e^{itx} g_N = \sum_{k=1}^{K} c_k \sum_{n=1}^{N_k} \int_{a_{k,n}}^{b_{k,n}} e^{itx}$$

but every term of this summation can be made arbitrarily small by part (i). In particular, this whole thing can be made smaller than 0.0001ε in absolute value. Observe $0.0001\varepsilon + 0.0001\varepsilon + 0.0001\varepsilon \leq \varepsilon$. Win.

Solution 4

Problem 4.1 — We say that $f \in L^2(\mathbb{R})$ is L^2-differentiable with L^2-derivative Df if $\|\tau_h f - f - hDf\|_2 / h \to 0$ as $h \to 0$, where $\tau_h f(x) = f(x+h)$. Show that the function $f(x) = \max(1 - |x|, 0)$ is L^2-differentiable and find its L^2-derivative.

Well, we're basically allowed to ignore non-differentiable sets of measure 0. The L^2 derivative is

$$Df = \begin{cases} 0 & x < -1 \\ 1 & -1 \le x < 0 \\ -1 & 0 \le x < 1 \\ 0 & x \ge 1 \end{cases}$$

such that for all $|h| < 1$,

$$\tau_h f - f - hDf = \begin{cases} 0 & x < -1-h \\ x+1+h & -1-h \le x < -1 \\ 0 & -1 \le x < -h \\ -2x-2h & -h \le x < 0 \\ 0 & 0 \le x < 1-h \\ x-1+h & 1-h \le x < 1 \\ 0 & x \ge 1 \end{cases}$$

Okay, the squared-L^2 norm of this thing (all parts integrated with usual Riemann integral) is

$$\frac{h^3}{3} + 4 \cdot \frac{h^3}{3} + \frac{h^3}{3} = 2h^3$$

Thus the L^2 norm is $\sqrt{2}h^{3/2}$, and the limit is of $\sqrt{2}h^{1/2}$, which vanishes as $h \to 0$. Done.

Problem 4.2 — Suppose that $f \in L^1 \cap L^2$ is L^2-differentiable. Show that $u\hat{f}(u) \in L^2$.

Note that $Df \in L^2$, because otherwise the above integral is always ∞.

Well... let's just do it! Rewrite:

$$\frac{\tau_h f - f}{h} \overset{\|\bullet\|_2}{\to} Df$$

(as functions in u). We know that the Fourier transform (or rather, its extension from $\mathcal{L}^1 \cap \mathcal{L}^2$ to $\mathcal{L}^2$) is a Hilbert space isomorphism (theorem 7.6.1), so it preserves limits in the L^2 norm. Thus,

$$\int_{\mathbb{R}} \frac{\tau_h f(x) - f(x)}{h} e^{iux} dx \overset{\|\bullet\|_2}{\to} [\widehat{Df}]$$

as a function in u.

Idea (Sanity Check) — As the main point of this problem is that $[\widehat{Df}] \in \mathcal{L}^2$, let me talk about it.

Here, we are defining $[\widehat{Df}]$ as described in 7.6.1; it is the limit of the fourier transforms of a sequence of functions in $\mathcal{L}^1 \cap \mathcal{L}^2$ which converge in $\mathcal{L}^2$. As the Plancherel identity holds for all functions in $\mathcal{L}^1 \cap \mathcal{L}^2$, we know that this limiting

function is also in $\mathcal{L}^2$.

However, observe that we can explicitly compute the LHS:

$$\begin{aligned}\int_{\mathbb{R}} \frac{\tau_h f(x) - f(x)}{h} e^{iux} dx &= \int_{\mathbb{R}} f(x) e^{iux} \frac{e^{-ihu} - 1}{h} dx \\ &= \frac{e^{-ihu} - 1}{h} \hat{f}(u) \\ &\to -iu\hat{f}(u)\end{aligned}$$

pointwise; this works *literally everywhere*. Convergence in L^2 implies that there exists a subsequence which converges pointwise almost everywhere. Thus, $[\widehat{Df}] = [-iu\hat{f}]$, and $u\hat{f} \in L^2$.

Problem 4.3 — Deduce that f has a continuous version and that $\|f\|_\infty \leq C \left\|(1+|u|)\hat{f}(u)\right\|_2$ for some constant $C < \infty$, to be determined.

Well,

$$[\hat{f}] = \left[\frac{i}{u}\widehat{Df}\right]$$

We know that $\widehat{Df} \in L^2$, and $\hat{f} \leq \|f\|_1 < \infty$, thus

$$\begin{aligned}\int_{\mathbb{R}} \left|\hat{f}(u)\right| du &= \int_{|u|\leq 1} \left|\hat{f}(u)\right| du + \int_{|u|\geq 1} \left|\frac{\widehat{Df}(u)}{u}\right| du \\ &\leq \sqrt{\int_{|u|\leq 1} \left|\hat{f}(u)\right|^2} \sqrt{\int_{|u|\leq 1} 1} + \sqrt{\int_{|u|\geq 1} \frac{1}{u^2} du} \cdot \sqrt{\int_{|u|\geq 1} \left|\widehat{Df}(u)\right|^2 du} \\ &\leq \sqrt{2}\left\|\hat{f}\right\|_2 + \sqrt{2}\left\|\widehat{Df}\right\|_2 \\ &\leq \sqrt{2}\left\|\max \hat{f}, \widehat{Df}\right\|_2 + \sqrt{2}\left\|\max \hat{f}, \widehat{Df}\right\|_2 \\ &\leq 2\sqrt{2}\left\|(1+|u|)\hat{f}(u)\right\|_2\end{aligned}$$

Hence $\hat{f} \in L^1$, and the Fourier inversion theorem applies; f is a version of the inverse fourier transform of $\hat{f}$.

Thus, f has a continuous version, because the inverse fourier transform is continuous (for the same reasons as the fourier transform) and $\|f\|_\infty \leq \left\|\hat{f}\right\|_1 / (2\pi)$ (for the same reasons as the fourier transform). So you can pick $C = 100000$ or whatever and we're done.

Solution 5 Portmanteau Theorem

Problem 5.1 — Let $(X_n : n \in \mathbb{N})$ be a sequence of random variables in $\mathbb{R}$ and let X be another random variable. Show that $X_n \to X$ weakly if and only if $X_n \to X$ in distribution.

This theorem is famous, important, and also completely tedious.

Nobody cares about your random variables, let μ_n and μ be their laws and we'll talk about those.

5.1 Upgrading to Metric spaces

Aiya first let me upgrade "convergence in distribution" to a more general form valid on metric spaces called "convergence in pineapples" so I can work with morally correct objects. The overarching claim for this section:

Claim (Convergence in distribution $\iff$ Convergence in pineapples) — Recall that a sequence of probability measure μ_n on $\mathbb{R}$ endowed with cumulative distribution function F_n **converges in distribution** to μ if

$$F_n(x) \to F(x)$$

for all x such that F is continuous.

On an arbitrary metric space X, a sequence of probability measures μ_n on X **converges in pineapples** to μ if for all open sets $U \subset X$,

$$\liminf \mu_n(U) \geq \mu(U)$$

Then, convergence in distribution $\iff$ convergence in pineapples.

First, let me delete the stupid CDFs from the definition. We know $F(x) = \mu((-\infty, x])$, and this is clearly already continuous from the right (by continuity of measure from above in probability spaces).

Claim — I claim that $F(x)$ is continuous at x iff $\mu(\{x\}) = 0$.

Proof. The lower limit is

$$\lim_{\varepsilon \to 0} F(x - \varepsilon) = \lim_{\varepsilon \to 0} \mu((-\infty, x - \varepsilon]) = \mu((-\infty, x)) = F(x) - \mu(\{x\})$$

by continuity of measure from below. Thus the lower limit coincides with the upper limit of $F(x)$ iff $\mu(\{x\}) = 0$. □

Great. Let's show distribution $\implies$ pineapples now.

Recall that any open set $U \subset \mathbb{R}$ is a *countable* disjoint union of open intervals (a, b), and thus the problem reduces to the case of a single open interval (a, b).

But this is basically clear... here are four cases.

Claim (Limit Interchange) — Suppose you have a double sequence of real numbers $0 \leq a_{mn} \leq 1$. Suppose also that a_{mn} is non-decreasing in m and non-decreasing in

n; for *all* $m \leq m'$ and $n \leq n'$, we have $a_{mn} \leq a_{m'n'}$. Then, we can swap the limits:

$$\lim_{n\to\infty} \lim_{m\to\infty} a_{mn} = \lim_{m\to\infty} \lim_{n\to\infty} a_{mn}$$

Proof.

$$\begin{aligned}
\lim_{n\to\infty} \lim_{m\to\infty} a_{mn} &= \lim_{n\to\infty} \sup_m a_{mn} \\
&= \sup_n \sup_m a_{mn} \\
&= \sup_m \sup_n a_{mn} \\
&= \sup_m \lim_{n\to\infty} a_{mn} \\
&= \lim_{m\to\infty} \lim_{n\to\infty} a_{mn}
\end{aligned}$$

We were able to set $\lim_n \sup_m = \sup_n \sup_m$ because supremums preserve inequalities, and thus $\sup_m a_{mn}$ is increasing in n. □

Now, suppose $\mu_n \to \mu$ weakly. Pick some a such that $F(x)$ is continuous. The first order of business is to model two functions: $\mathbb{1}_{a<x}$, and $\mathbb{1}_{a\neq x}$. Let:

- f_n be 1 on $x \leq a - \frac{1}{n}$, 0 on $x \geq a$, and linear in between.
- $g_n = \min(n\,|x-a|\,, 1)$; this forms a V-shape near a.

Observe that f_n and g_n are continuous functions bounded between 0 and 1 which are increasing and converges pointwise to $\mathbb{1}_{x\leq a}$ and $\mathbb{1}_{x\neq a}$, respectively.

Idea — It might seem odd that we chose to model these instead of $\mathbb{1}_{a\leq x}$ and $\mathbb{1}_{a=x}$, but I'm pretty sure we need to pass through the limit interchange lemma above, and I can only do this when things are increasing.

Seeing as this is a rather technical problem, let me justify each step.

$$\begin{aligned}
F(a) - \mu(\{a\}) &= \int \mathbb{1}_{x<a} d\mu && \text{(by definition of } F\text{)} \\
&= \int \lim_{n\to\infty} f_n d\mu && \text{(by definition of } f_n\text{)} \\
&= \lim_{n\to\infty} \int f_n d\mu && \text{(by monotone convergence theorem)} \\
&= \lim_{n\to\infty} \lim_{m\to\infty} \int f_n d\mu_m && \text{(by definition of weak convergence)} \\
&= \lim_{m\to\infty} \lim_{n\to\infty} \int f_n d\mu_m && \text{(by limit interchange)} \\
&= \lim_{m\to\infty} \int \lim_{n\to\infty} f_n d\mu_m && \text{(by monotone convergence theorem)} \\
&= \lim_{m\to\infty} \int \mathbb{1}_{x<a} d\mu_m && \text{(by definition of } f_n\text{)} \\
&= \lim_{m\to\infty} F_m(a) - \mu_m(\{a\}) && \text{(by definition of } F_m\text{)}
\end{aligned}$$

Running this exact same calculation on g_n shows

$$1 - \mu_m(\{a\}) = \lim_{m\to\infty} 1 - \mu_m(\{a\})$$

Subtracting these two results, we find that $F_m(a) \to F(a)$ for all a such that F is continuous. This concludes.

> **Idea —** Sheesh, thread the needle.

Now, suppose $X_n \to X$ in distribution. Give me a continuous bounded function f. I'm going to approximate f by a series of piecewise constant functions, using standard real analysis.

> (Constructing f_n) Here's how we construct function f_n. First, find some M_n such that $\mu(|X| \geq M_n) < \frac{1}{n\|f\|_\infty}$ (this is possible because μ is a probability measure).
>
> Set $f_n = 0$ outside $[-M_n, M_n]$. Second, on $[-M_n, M_n]$, f is continuous, and thus uniformly continuous (by standard results). Let $\varepsilon = \frac{1}{n}$.
>
> - f is UC, thus $\forall\varepsilon\exists\delta$ such that if $|x - y| < \delta$, then $|f(x) - f(y)| < \varepsilon$.
> - Use δ as a discretization; split $[a, b]$ into a finite number of intervals of length at most δ, taking care to avoid a countable number of points which will be specified shortly.
> - In each interval I_i, set $f_k(I_i)$ to be an arbtrarily chosen value from $f(I_i)$.
> - In particular, f_k is uniformly bounded by the same bounds as f.
>
> Then, $|f_n| < \|f\|_\infty$ everywhere, and $|f(x) - f_n(x)| < \varepsilon$ everywhere on $[-M_n, M_n]$.

Finally, give me some ε, and I'll find you an n such that $|\mu(f) - \mu_m(f)| < \varepsilon$ for all $m > n$.

First, let $k > \frac{0.0001}{\varepsilon}$. Find n such that for all $m > n$,

- $\mu_m((-\infty, M_k])$ is within $0.0001\varepsilon/\|f\|_\infty$ of $\mu((-\infty, M_k])$.
- $\mu_m((-\infty, -M_k])$ is within $0.0001\varepsilon/\|f\|_\infty$ of $\mu((-\infty, -M_k])$.
- $\mu_m(f_k) - \mu(f_k) < 0.0001\varepsilon$.

This is possible because $\mu_m \to \mu$ weakly. (The last line is true because $\mu_m(f_k)$ is just the measure of a bunch of intervals).

> **Idea** (Detail) **—** Okay, we only know $\mu_m((-\infty, x]) \to \mu((-\infty, x])$ for all x such that F is continuous.
>
> However:
>
> - F is discontinuous at x iff $\mu(\{x\}) > 0$.
> - Suppose there existed an uncountable set S such that $\mu(\{x\}) > 0$ for all $x \in S$. Then, one can choose a finite subset of S with arbitrarily large measure (by splitting S into buckets $(\frac{1}{n+1}, \frac{1}{n})$ and finding a bucket with uncountably many things). This is a contradiction, thus the set of discontinuities is countable.
> - But $\mathbb{R}$ is uncountable.

> So the upshot is that if M_k fails, you can just poke it to a slightly larger value for which F is actually continuous, and ignore this problem.
>
> And actually, the number of discontinous points for all F_m is countable too. So in our definition of f_k above, let's just pick our I_i's to avoid all of these points.

Then, for all $m > n$,

$$\begin{aligned}
|\mu_m(f) - \mu_m(f_k)| &\leq \int |f - f_k| \, \mu_m(dx) \\
&\leq \int |f| \, \mathbb{1}\left((-\infty, -M_k) \cup (M_k, \infty)\right) \mu_m(dx) + \int 0.0001\varepsilon \mathbb{1}\left([-M_k, M_k]\right) \mu_m(dx) \\
&\leq \|f\|_\infty \, \mu_m((-\infty, -M_k] \cup [M_k, \infty)) + 0.0001\varepsilon \\
&\leq \|f\|_\infty \, \mu((-\infty, -M_k] \cup [M_k, \infty)) + \|f\|_\infty \cdot \frac{0.0001\varepsilon}{\|f\|_\infty} + 0.0001\varepsilon \\
&\leq 0.0001\varepsilon + 0.0001\varepsilon + 0.0001\varepsilon \\
&\leq 0.0003\varepsilon
\end{aligned}$$

Similarly,

$$\begin{aligned}
|\mu(f) - \mu(f_k)| &\leq \int |f - f_k| \, \mu(dx) \\
&\leq \int |f| \, \mathbb{1}\left((-\infty, -M_k) \cup (M_k, \infty)\right) \mu(dx) + \int 0.0001\varepsilon \mathbb{1}\left([-M_k, M_k]\right) \mu(dx) \\
&\leq \|f\|_\infty \, \mu((-\infty, -M_k] \cup [M_k, \infty)) + 0.0001\varepsilon \\
&\leq 0.0001\varepsilon + 0.0001\varepsilon \\
&\leq 0.0002\varepsilon
\end{aligned}$$

Finally, the difference between $\mu_m(f_k)$ and $\mu(f_k)$ is at most 0.0001ε from before. So in total $|\mu_m(f) - \mu(f)| < 0.0006\varepsilon$. Done.

Solution 6

Problem 6.1 — Let $X_1, \ldots, X_n$ be independent $N(0,1)$ random variables. Show that

$$\left(\overline{X}, \sum_{m=1}^{n} (X_m - \overline{X})^2\right) \text{ and } \left(\frac{X_n}{\sqrt{n}}, \sum_{m=1}^{n-1} X_m^2\right) \tag{$\star$}$$

have the same distribution, where $\overline{X} = (X_1 + \cdots + X_n)/n$.

First of all, here's a completely trivial fact:

Claim — If $X, Y : \Omega \to E$ are two random variables which have the same distribution and $f : E \to F$ is a measurable function, then $f(X)$ and $f(Y)$ have the same distribution.

Proof. The definition of having the same distribution is that

$$\mu \circ X^{-1} = \mu \circ Y^{-1}$$

Then clearly

$$\mu \circ (f \circ X)^{-1} = \mu \circ X^{-1} \circ f^{-1} = \mu \circ Y^{-1} \circ f^{-1} = \mu \circ (f \circ Y)^{-1}$$

□

Now, let me write down an orthogonal matrix U. Actually, I'm only going to fill in the bottom row:

$$U = \begin{pmatrix} ? & \cdots & ? \\ \vdots & \ddots & \vdots \\ ? & \cdots & ? \\ \frac{1}{\sqrt{n}} & \cdots & \frac{1}{\sqrt{n}} \end{pmatrix}$$

and by e.g. Gram-Schmidt, I can fill in the rest of the matrix with an orthonormal basis for the orthogonal complement of $\vec{\mathbf{a}} = \frac{1}{\sqrt{n}}(1, \ldots, 1)^\top$.

Great! Consider the function

$$f : \mathbb{R}^n \to \mathbb{R}^2 : \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \to \begin{pmatrix} \frac{x_n}{\sqrt{n}} \\ \|x\|_2^2 - x_n^2 \end{pmatrix}$$

It is clear that the RHS of $\star$ is $f(X)$ where $X \in \mathbb{R}^n = (X_1, \ldots, X_n)^\top$. I claim that the LHS is equal to $f(UX)$, where UX is a completely legal thing to write down (I am composing $U \circ X$ where $U : \mathbb{R}^n \to \mathbb{R}^n$).

It suffices to show that (as linear algebra identities)

$$\sum_{m=1}^{n} (X_m - \overline{X})^2 = \|UX\|_2^2 - (UX)_n^2 = \|X\|_2^2 - n\overline{X}^2$$

This is basically obvious and left as an exercise. No I'm kidding, uh, the LHS is

$$\sum X_i^2 - 2\overline{X} \sum X_i + \overline{X}^2 \cdot n = \|X\|_2^2 - n\overline{X}^2$$

lol.

By the claim, it suffices to show that UX and X have the same distribution. But if you stare at their distributions, this is completely obvious:

$$f_{UX}(\vec{\mathbf{x}}) = \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2}\|U\vec{\mathbf{x}}\|_2^2} = \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2}\|\vec{\mathbf{x}}\|_2^2} = f_X(\vec{\mathbf{x}})$$

Solution 7

Problem 7.1 — Let $(E, \mathcal{E}, \mu)$ be a measure space and $\theta : E \to E$ a measure-preserving transformation. Show that $\mathcal{E}_\theta = \{A \in \mathcal{E} : \theta^{-1}(A) = A\}$ is a σ-algebra.

$E \subset \mathcal{E}_\theta$ because $\theta^{-1}(E) = E$. Suppose $A \in \mathcal{E}_\theta$; then

$$\theta^{-1}(E \setminus A) = \theta^{-1}(E) \setminus \theta^{-1}(A) = E \setminus A \implies E \setminus A \in \mathcal{E}_\theta.$$

Finally, if $A_1, \dots A_n \in \mathcal{E}_\theta$, then

$$\theta^{-1}\left(\bigcup_i A_i\right) = \bigcup_i \theta^{-1}(A_i) = \bigcup_i A_i$$

as desired.

Problem 7.2 — Show that a measurable function f is $\mathcal{E}_\theta$-measurable if and only if it is invariant, that is $f \circ \theta = f$.

I will assume f points into $\mathbb{R}$. Otherwise, there are counterexamples; the measurability condition can't distinguish between points if the measure space isn't "Hausdorff".

Suppose f is $\mathcal{E}_\theta$-measurable. Then, noting that points are measurable in $\mathbb{R}$, $f^{-1}(x)$ is measurable in $\mathcal{E}_\theta$. This implies $\theta^{-1}(f^{-1}(x)) = f^{-1}(x)$ for all x. Now, give me any $x \in E$. I know that $x \in \theta^{-1}(f^{-1}(f(x)))$. Thus,

$$f(\theta(x)) = f(f^{-1}(f(x))) = f(x)$$

On the other hand, suppose $f \circ \theta = f$. Then, for all measurable sets $A \subset \mathbb{R}$, $f^{-1}(A) = \theta^{-1}(f^{-1}(A))$, i.e. $f^{-1}(A) \in \mathcal{E}_\theta$.

Solution 8

Problem 8.1 — Show that, if θ is an ergodic measure-preserving transformation and f is a θ-invariant function, then there exists a constant $c \in \mathbb{R}$ such that $f = c$ a.e.

If E has measure 0, the problem is stupid, because all functions are equal almost everywhere. Suppose otherwise.

Observe that for any measurable $A \subset \mathbb{R}$, $f^{-1}(A)$ is θ-invariant; by ergodicity, either $f^{-1}(A)$ or $E \setminus f^{-1}(A)$ has measure 0.

Now, consider the sets $E_i = f^{-1}([i, i+1))$ for $i \in \mathbb{Z}$; this is a countable set, thus they cannot *all* have measure 0. Two of them cannot have nonzero measure, else neither of their complements would have measure 0. Thus, exactly one of them has nonzero measure; call it $k = i$.

Then, consider the dyadic blocks

$$E_{n,a} = \left[\frac{a}{2^n} + k, \frac{a+1}{2^n} + k\right), \qquad a = 0, \ldots, 2^n - 1$$

These have the property that for all n, they partition $[k, k+1)$. Thus, by the above reasoning, for each n there is exactly one a such that $E_{n,a}$ has nonzero measure. Call this a_n. Clearly $a_{n+1} \in \{2a_n, 2a_n + 1\}$; any other choice would contradict the fact that $E \setminus E_{n,a}$ has zero measure.

Thus, we find ourselves with a descending series of blocks

$$E_{0,a_0} \supset E_{1,a_1} \supset \ldots$$

such that $E \setminus E_{k,a_k}$ has measure 0. By continuity of measure from below,

$$\mu\left(E \setminus \inf E_{k,a_k}\right) = \lim_{k \to \infty} \mu\left(E \setminus E_{k,a_k}\right) = 0$$

But clearly $\inf E_{k,a_k}$ is some singleton $\{c\}$ (by definition of the real numbers). Thus, $E \setminus f^{-1}(\{c\})$ has measure 0, and we are done.